

Which Test Answers the Question Asked?

AP Statistics · Unit 3 · Topic 3.5 · TEACHER KEY

CED TETHER

- **Enduring Understanding VAR-6** — The normal distribution may be used to model variation.
- **LO VAR-6.D** — Identify the null and alternative hypotheses for a population proportion. [Skill 1.F]
- **VAR-6.D.1** — The null hypothesis is the situation that is assumed to be correct unless evidence suggests otherwise, and the alternative hypothesis is the situation for which evidence is being collected.
- **VAR-6.D.2** — For hypotheses about parameters, the null hypothesis contains an equality reference ($=$, $\geq$, or $\leq$), while the alternative hypothesis contains a strict inequality ($<$, $>$, or $\neq$). The type of inequality in the alternative hypothesis is based on the question of interest. Alternative hypotheses with $<$ or $>$ are called one-sided, and alternative hypotheses with $\neq$ are called two-sided. Although the null hypothesis for a one-sided test may include an inequality symbol, it is still tested at the boundary of equality.
- **VAR-6.D.3** — The null hypothesis for a population proportion is: $H_0: p = p_0$, where p_0 is the null hypothesized value for the population proportion.
- **VAR-6.D.4** — A one-sided alternative hypothesis for a proportion is either $H_a: p < p_0$ or $H_a: p > p_0$. A two-sided alternative hypothesis is $H_a: p \neq p_0$.
- **VAR-6.D.5** — For a one-sample z-test for a population proportion, the null hypothesis specifies a value for the population proportion, usually one indicating no difference or effect.
- **LO VAR-6.E** — Identify an appropriate testing method for a population proportion. [Skill 1.E]
- **VAR-6.E.1** — For a single categorical variable, the appropriate testing method for a population proportion is a one-sample z-test for a population proportion.
- **LO VAR-6.F** — Verify the conditions for making statistical inferences when testing a population proportion. [Skill 4.C]
- **VAR-6.F.1** — In order to make statistical inferences when testing a population proportion, we must check for independence and that the sampling distribution is approximately normal:
 - a. To check for independence:
 - i. Data should be collected using a random sample or a randomized experiment.
 - ii. When sampling without replacement, check that $n \leq 10\%N$.
 - b. To check that the sampling distribution of $\hat{p}$ is approximately normal (shape):
 - i. Assuming that H_0 is true ($p = p_0$), verify that both np_0 and $n(1 - p_0)$ are at least 10 so that the sample size is large enough to support an assumption of normality.

Essential question: How do the research direction and null model determine a valid one-proportion test setup?

DOK-3 skill: 4.C · **FRQ pattern:** directional-question-null-based-conditions-which-test-setup

DOK rationale: Part (c) adjudicates two plausible test setups by coordinating the research direction, parameter-versus-statistic distinction, procedure choice, and null-based condition checks, then explains what each mismatch would test instead.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- What do p and $84/180$ each describe?

- Would a lower current sample proportion count as evidence that ridership increased?
- Does large counts model the null or the observed sample?

[WARN] LOOK FOR

- Choosing two-sided merely because samples vary both ways.
- Putting the sample proportion in a hypothesis.
- Using observed rather than null-expected counts.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Chooses Maya and ties the greater-than alternative to the pre-stated increase.
- **judgment-2** (required) — Explains that 72 and 108 are expected under the null benchmark $p=0.40$, whereas 84 and 96 are observed counts.
- **judgment-3** (required) — Explains that Theo's two-sided alternative would treat a decrease as evidence of change.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

[STAR] THE PROBLEM — annotated

A district of 2,400 students had bus ridership $p_0 = 0.40$ before a new route. Before collecting data, the director asks whether the current proportion p has *increased*. An SRS of 180 current students finds 84 riders.

Maya: “Use $H_0 : p = 0.40$ versus $H_a : p > 0.40$ and a one-sample z -test. Check large counts with 72 and 108.”

Theo: “Use $H_a : p \neq 0.40$ because samples can differ either way. Check large counts with the observed 84 and 96.”

Current SRS counts

Group	Bus	Other	Total
Students	84	96	180

FIRST TAKE — one sentence before you work the parts

Does the director want to detect any change, or only more bus riders? Use a phrase from the study.

Key: Not graded. Separate the question asked from the sample observed.

[BOOK] SET UP THE QUESTION BEFORE USING THE SAMPLE (keep on the page)

Define p as the **population proportion**. For a benchmark p_0 , use $H_0 : p = p_0$. Choose $H_a : p > p_0$, $p < p_0$, or $p \neq p_0$ from the research question stated *before* seeing data; $\hat{p}$ is sample evidence and never belongs in the hypotheses. For one categorical variable and one sample, use a one-sample z -test for a population proportion. Verify random data, $n \leq 0.10N$ when sampling without replacement, and the null-expected counts $np_0 \geq 10$ and $n(1 - p_0) \geq 10$.

DOK 1 (a) Define the population parameter p in context and identify the null value p_0 .

Key: Here p is the current proportion of all 2,400 district students who use the school bus. The null value is the pre-route benchmark $p_0 = 0.40$.

DOK 2 (b) Compute $\hat{p}$, name the testing procedure, and verify the random, 10 percent, and large-counts conditions using the appropriate values.

Key: $\hat{p} = 84/180 = 7/15 \approx 0.4667$. Use a one-sample z -test for p . The SRS is random; $180 \leq 0.10(2,400) = 240$; and under H_0 , $180(0.40) = 72$ and $180(0.60) = 108$ are at least 10. All conditions hold.

DOK 3 (c) In 3–4 sentences, choose a setup using the director’s word increased. Explain why the count check uses 72 and 108 rather than 84 and 96. State what different question Theo’s alternative would answer.

Key: Maya is warranted. The pre-data word *increased* requires $H_a : p > 0.40$; $p_0 = 0.40$ is the benchmark, while $\hat{p} = 84/180$ is evidence, not a hypothesis value. Large counts models H_0 , so it uses expected counts 72 and 108. Theo’s two-sided test asks about any change and would count a decrease as evidence. His observed 84 and 96 happen to exceed 10, but using them could make validity change with the sample and mislead readers about the null sampling distribution.